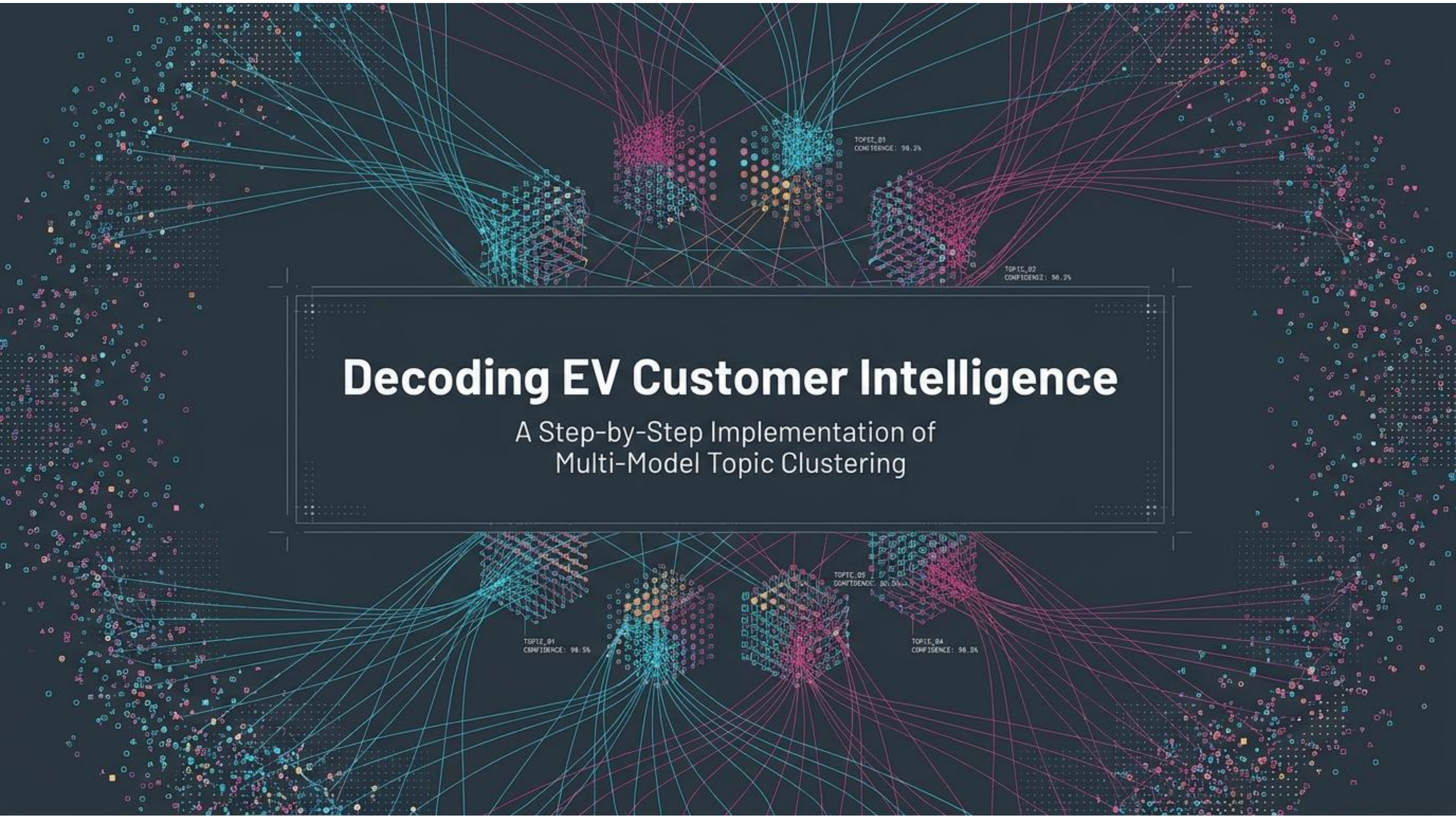




Decoding EV Customer Intelligence

A Step-by-Step Implementation of
Multi-Model Topic Clustering

TOPIC_01
CONFIDENCE: 98.2%

TOPIC_02
CONFIDENCE: 96.2%

TOPIC_03
CONFIDENCE: 98.5%

TOPIC_04
CONFIDENCE: 97.8%

TOPIC_05
CONFIDENCE: 96.3%

Transforming Raw Feedback into a Strategic Taxonomy

We engineered a robust, automated Python pipeline to ingest, clean, and cluster unstructured customer reviews. By running three competing NLP algorithms in parallel, we eliminated model bias and synthesised hyper-specific customer pain points into a reliable taxonomy for product strategy.

1

Pipeline

A 7-step automated architecture from dataset generation to Excel synthesis.

3

NLP Engines

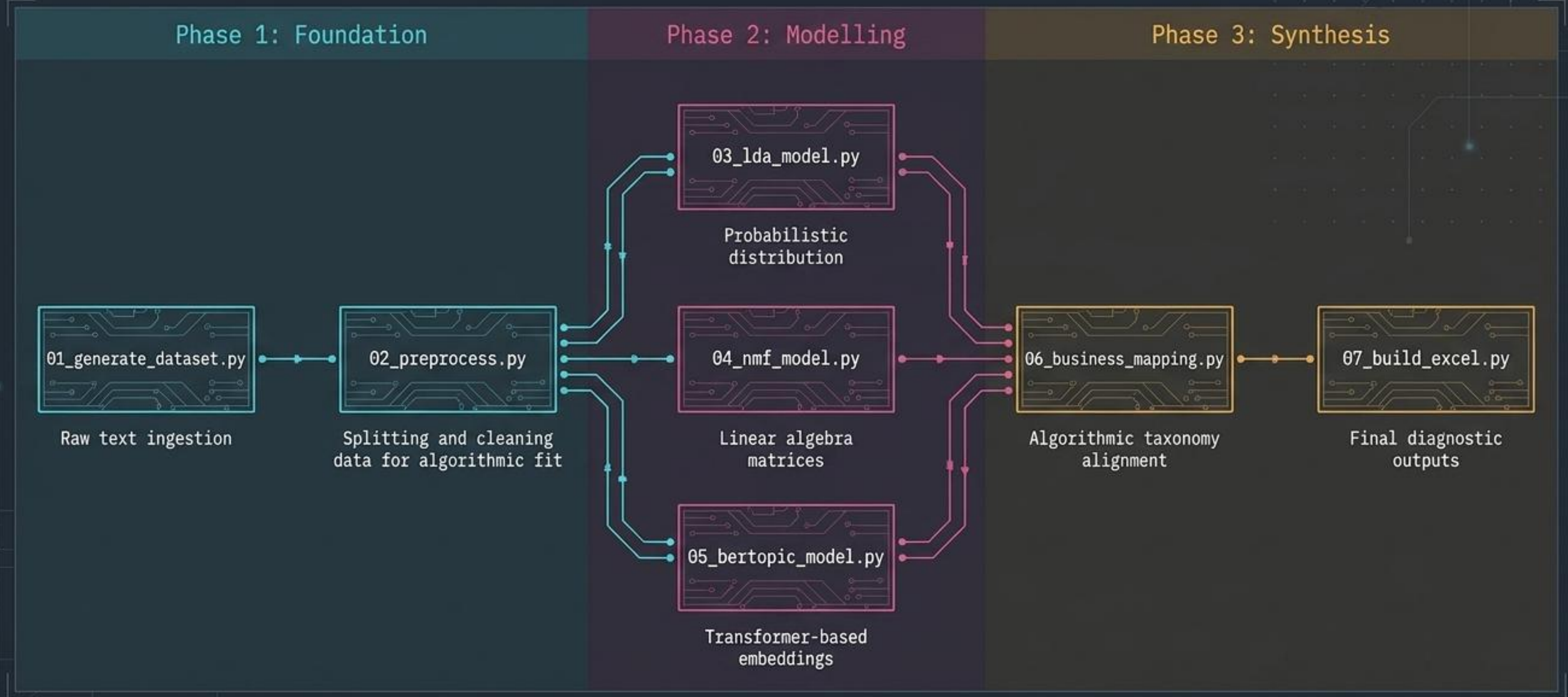
Utilising **LDA**, **NMF**, and **BERTopic** for cross-validation and micro-clustering.

10

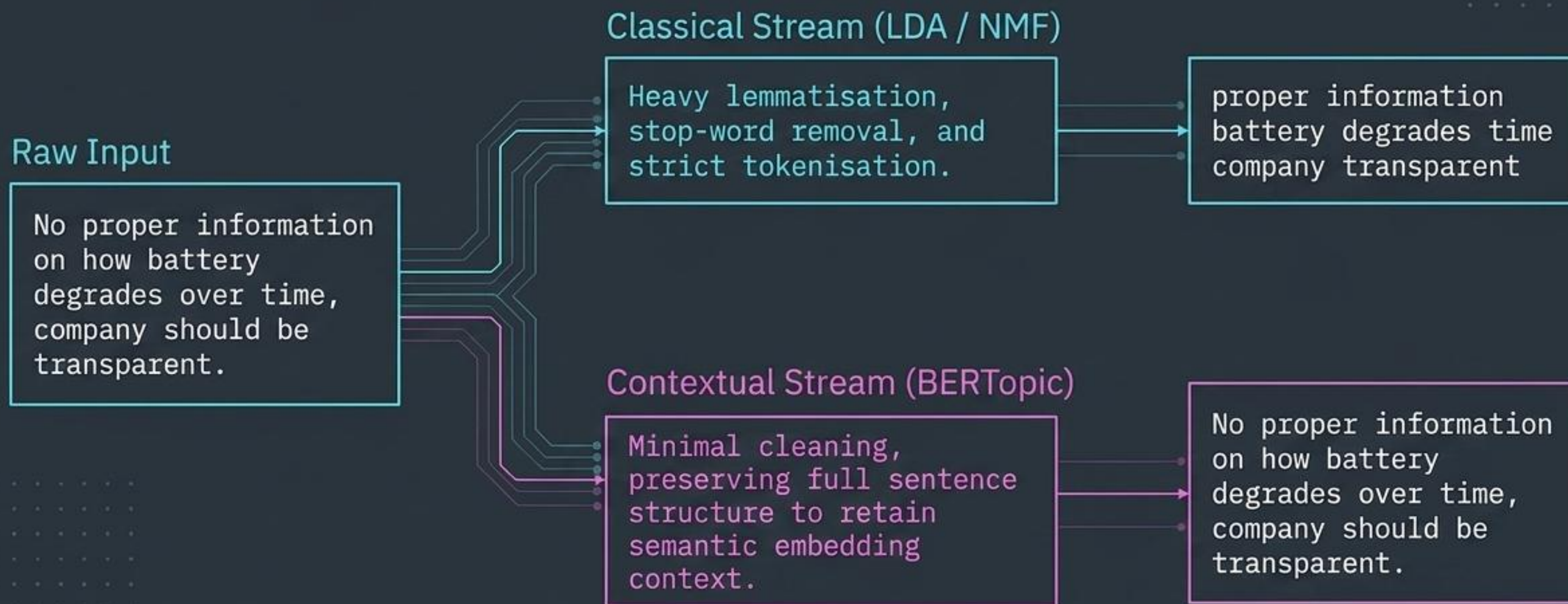
Business Themes

Distilling thousands of unstructured reviews into a unified product taxonomy.

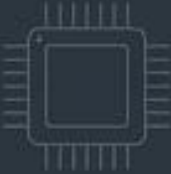
The End-to-End Analytics Architecture



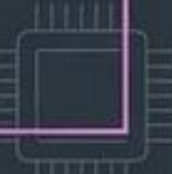
Diverging Preprocessing Streams Preserve Contextual Value



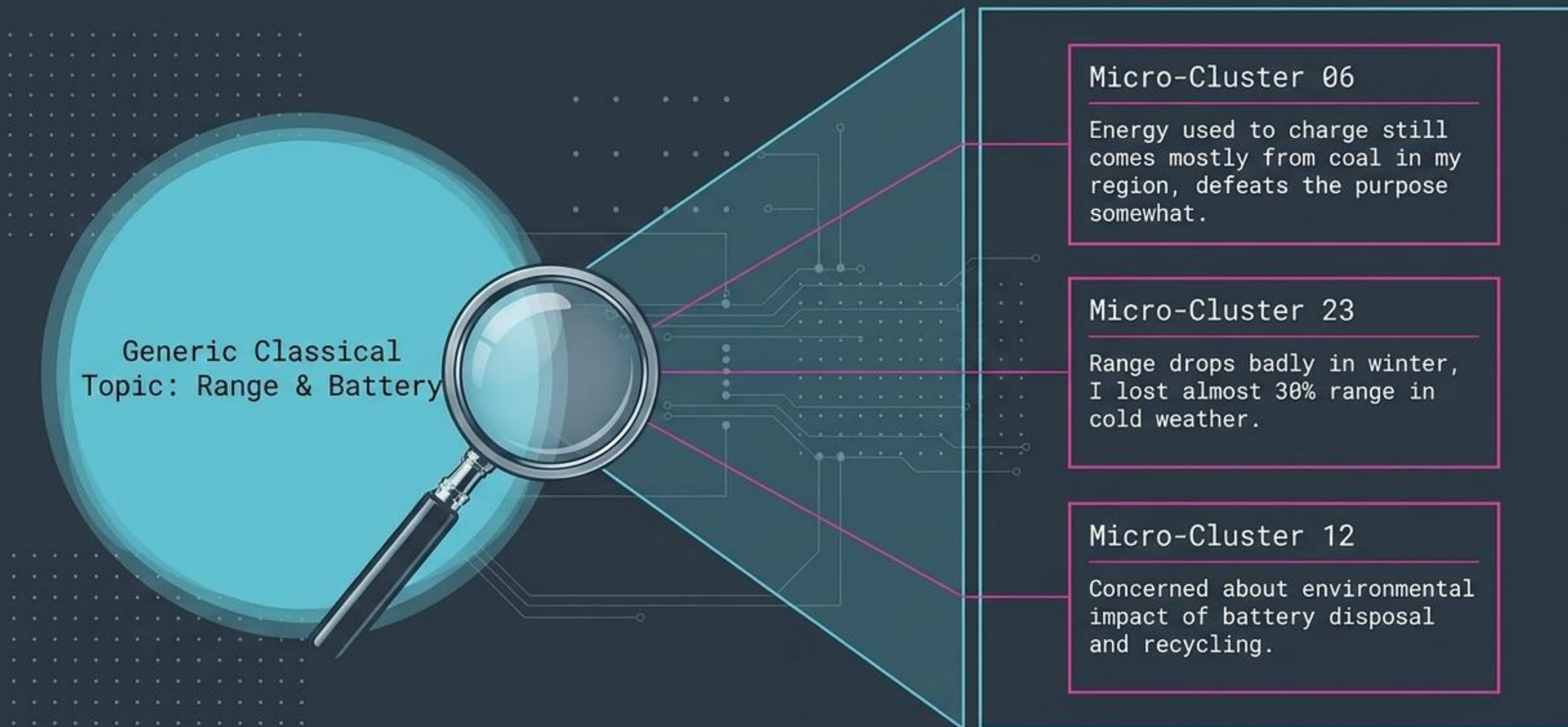
The Tri-Model Diagnostic Engine



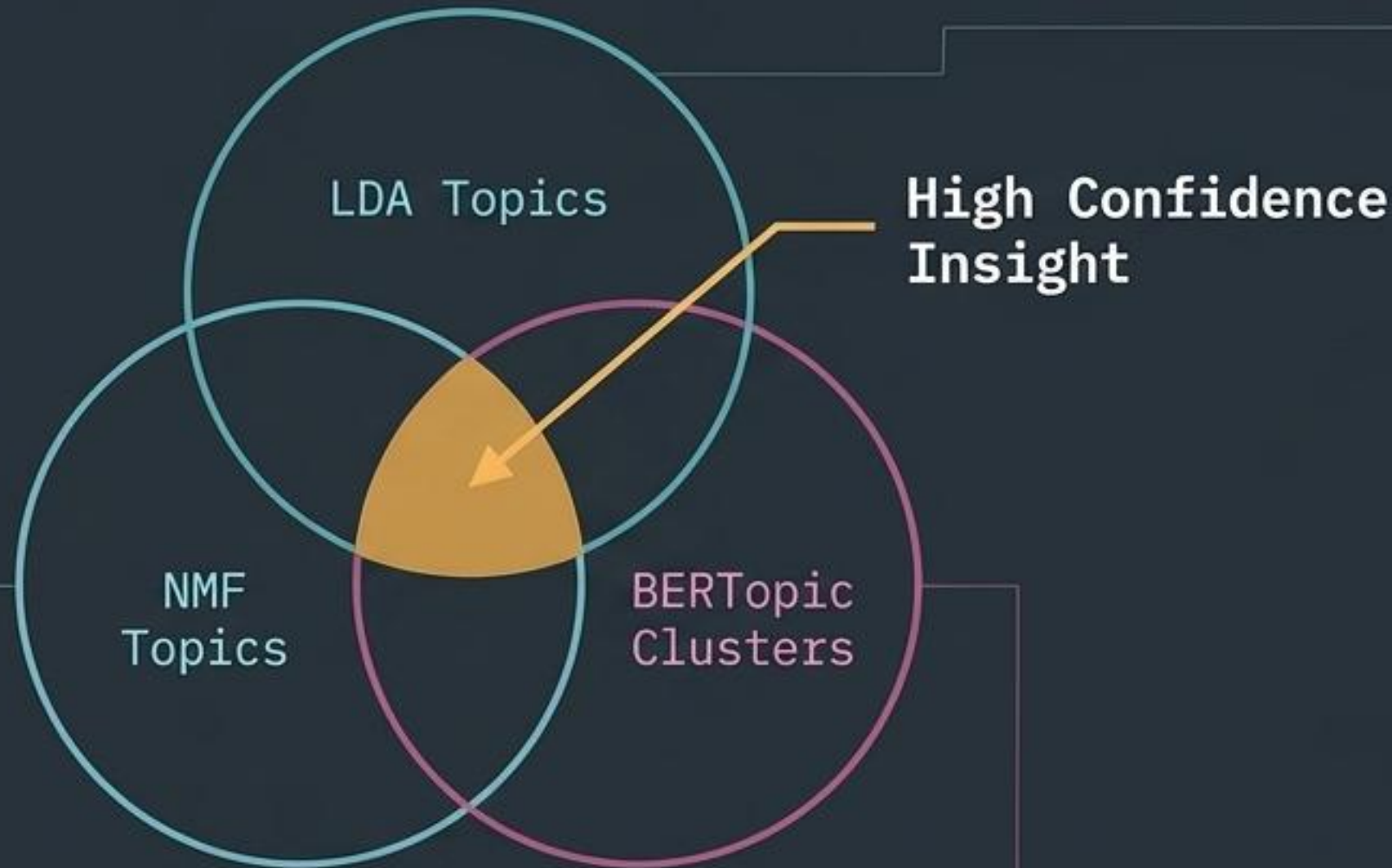
	LDA (Latent Dirichlet Allocation)	NMF (Non-Negative Matrix Factorisation)	BERTopic
Method	Probabilistic topic bucketing.	Linear algebra and latent feature co-occurrence.	Transformer-based language embeddings.
Strength	Identifies broad, distinct categories.	Surfaces latent operational patterns.	Highly granular, context-aware micro-clustering.
Dataset Example	Topic 4: software, month, year, battery, keep	Topic 8: service, service center, center, appointment long, center needed	Topic 12: battery disposal, environmental, disposal, disposal recycling



Granularity in Action: The BERTopic Advantage



The Consensus Engine Maps Algorithms to Strategy

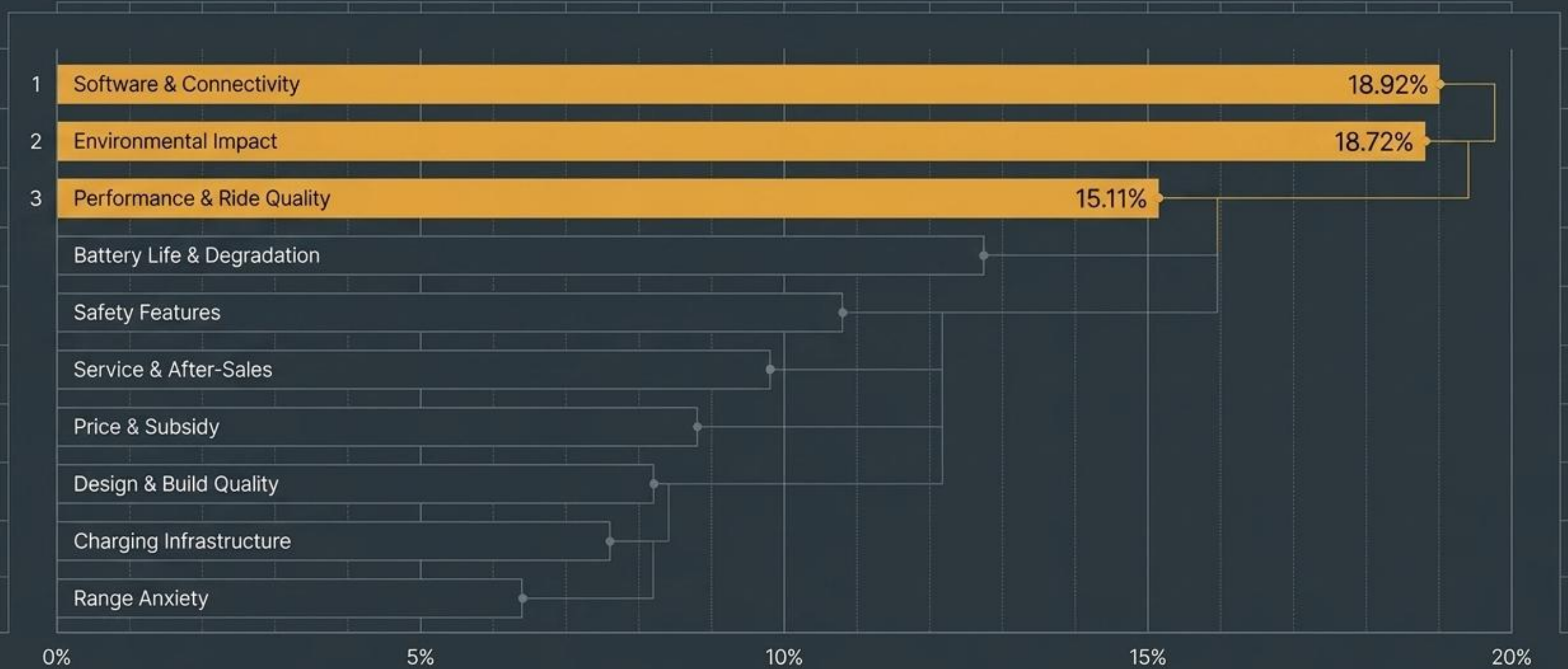


EV_Customer_Insights_Topic_Modeling.xlsx	
ev_customer_feedback_final.csv	
pain_points_vs_strengths.csv	
theme_prevalence_cross_model.csv	
theme_by_product_type.csv	
bertopic_document_topics.csv	
bertopic_topic_summary.csv	
ev_customer_feedback_clean.csv	
lda_topic_summary.csv	
ev_customer_feedback_raw.csv	
nmf_document_topics.csv	
nmf_topic_summary.csv	
lda_document_topics.csv	

Unified Taxonomy Validation: Each review is mapped to 1 of 10 business themes. We rely on cross-model validation logic (**models_agree_all3** and **models_agree_2of3**) to ensure strategic recommendations are mathematically backed, not anomalous.

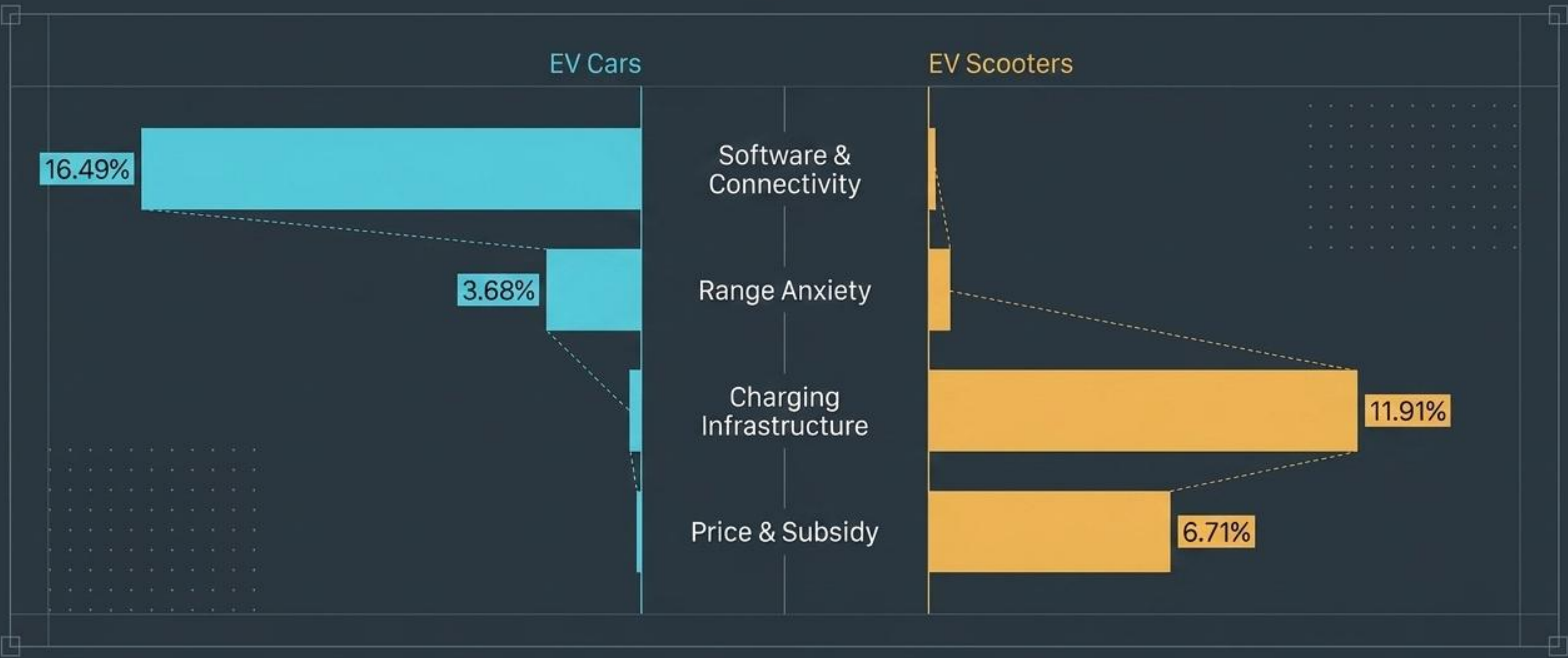
Insight I: The Cross-Model Theme Consensus

Surprisingly, software functionality and environmental anxiety outpace hardware performance as the dominant vectors of customer conversation.



Insight II: The Asymmetrical Priorities of Cars vs. Scooters

Takeaway: Car drivers view their vehicles as rolling computers constrained by range. Scooter riders view theirs as commuter appliances constrained by charging access and price.



Insight III: Isolating Critical Pain Points from Core Strengths



Strategic Directives: Translating Data to Decisions

Pillar 1: Overhaul Battery Degradation Transparency

The Data:

The lowest-rated pain point (3.76) combined with hyper-specific BERTopic anxiety regarding end-of-life disposal and winter range loss.

The Action:

Implement proactive dashboard battery health monitoring and clear, transparent warranty education.

Pillar 2: Double Down on Car Software Development

The Data:

Dominates EV Car feedback (16.49%).

The Action:

Transition from physical hardware marketing to viewing the EV Car primarily as a software-defined product. Fix app stability immediately.

Pillar 3: Subsidise Scooter Infrastructure

The Data:

Scooters drastically over-index on charging infrastructure complaints (11.91%).

The Action:

Shift capital from scooter range expansion into aggressive tier-2/tier-3 urban charging network partnerships.